



**SIMATS**  
**ENGINEERING**



**SIMATS**  
Saveetha Institute of Medical And Technical Sciences  
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# **SMART OBJECT RECOGNITION USING IMAGE ENHANCEMENT AND FEATURE INTELLIGENCE**

## **A CAPSTONE PROJECT REPORT**

*A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of*

**MLA0408 – DEEP LEARNING**

*to the award of the degree of*

**BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY**

**IN**

**ARTIFICIAL INTELLIGENCE & MACHINE LEARNING**

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## BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "SMART OBJECT RECOGNITION USING IMAGE ENHANCEMENT AND FEATURE INTELLIGENCE" has been carried out by **G. Naveen Kumar Reddy (192525288)**, **M. Pranai (192525068)** and **B. Bala Narayana (192525157)** under the supervision of **Dr. Sudhagar D** and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Artificial Intelligence & Machine Learning programme at Saveetha Institute of Medical and Technical Sciences, Chennai.

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**DECLARATION**

We, **G. Naveen Kumar Reddy [192525288]**, **M. Pranai [192525068]** and **B. Bala Narayana [192525157]** of the Department of Artificial Intelligence & Machine Learning, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "**Smart Object Recognition using Image Enhancement and Feature Intelligence**" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

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## SIMATS ENGINEERING

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### Individual Contribution Statement

*(Mandatory for all team projects)*

| Student Name / Register No.       | Specific Responsibilities                                     | Design & Development Contribution               | Testing & Analysis Contribution        | Report Contribution | Approx. Contribution (%) |
|-----------------------------------|---------------------------------------------------------------|-------------------------------------------------|----------------------------------------|---------------------|--------------------------|
| G. Naveen Kumar Reddy [192525288] | Problem formulation, image enhancement & preprocessing module | Enhancement and Harris corner detection modules | Accuracy and robustness testing        | Ch. 1–2             | 34%                      |
| M. Pranai [192525068]             | SIFT feature extraction, system integration, documentation    | SIFT module and system architecture             | Feature-matching and performance tests | Ch. 3–4, Appendices | 33%                      |
| B. Bala Narayana [192525157]      | PCA optimization, classification, visualization               | PCA and classification module                   | Comparison and evaluation tests        | Ch. 5, References   | 33%                      |

## ABSTRACT

Object recognition is a core computer vision task that underpins applications in surveillance, robotics, medical imaging, and autonomous systems. Recognition accuracy in practical settings is frequently degraded by noise, poor illumination, low contrast, and variation in scale, rotation, and viewpoint. This project presents an integrated pipeline for Smart Object Recognition that combines image enhancement, Harris corner-based feature detection, SIFT (Scale-Invariant Feature Transform) descriptor extraction and matching, and PCA (Principal Component Analysis)-based dimensionality reduction. Raw images are first pre-processed using noise reduction, contrast enhancement, sharpening, colour-space conversion, and normalization to improve clarity and feature visibility. The Harris Corner Detector then identifies stable, illumination- and rotation-invariant keypoints, which serve as anchor points for SIFT descriptor generation. The resulting high-dimensional SIFT descriptors are aggregated and projected onto a reduced set of principal components using PCA, retaining approximately 95% of the explained variance, before being passed to a classifier (SVM, KNN, or a neural network) for object recognition. The system is evaluated in terms of recognition accuracy, matching robustness, and computational efficiency under varying image conditions. The integrated pipeline demonstrates that combining enhancement, robust keypoint detection, invariant descriptors, and dimensionality reduction yields a recognition system that is more accurate and computationally efficient than approaches that use handcrafted features in isolation, while remaining suitable for near-real-time operation.

**Keywords:** Object recognition, image enhancement, Harris corner detection, SIFT, PCA, feature extraction, computer vision

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## LIST OF ABBREVIATIONS / SYMBOLS

| Abbreviation | Description                                      |
|--------------|--------------------------------------------------|
| SIFT         | Scale-Invariant Feature Transform                |
| PCA          | Principal Component Analysis                     |
| HE           | Histogram Equalization                           |
| CLAHE        | Contrast Limited Adaptive Histogram Equalization |
| DoG          | Difference of Gaussian                           |
| SVM          | Support Vector Machine                           |
| KNN          | K-Nearest Neighbours                             |
| FLANN        | Fast Library for Approximate Nearest Neighbours  |
| CNN          | Convolutional Neural Network                     |

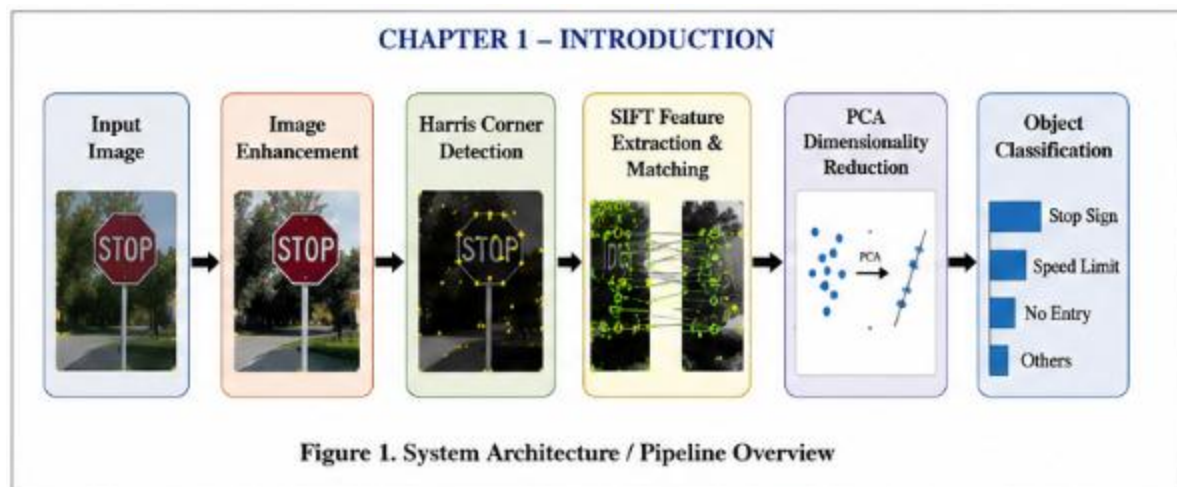
# CHAPTER 1

## INTRODUCTION AND ENGINEERING PROBLEM

### 1.1 Background

Object recognition is a fundamental problem in computer vision that combines image processing and machine learning techniques to identify and classify objects within images. Real-world images captured by cameras or sensors are rarely ideal: they often contain noise, poor illumination, low contrast, and geometric distortion, all of which reduce the reliability of downstream recognition. A recognition pipeline must therefore address image quality, extract features that are robust to scale, rotation, and illumination change, and manage the computational cost of high-dimensional feature representations.

This project proposes an integrated pipeline that combines Image Enhancement, Harris Corner Detection, SIFT Feature Extraction and Matching, and PCA-based dimensionality reduction to improve recognition accuracy while keeping computation tractable for near-real-time use.



### 1.2 Need for the Project

- Raw captured images often suffer from noise, poor illumination, and low contrast, which significantly reduce recognition accuracy.
- Traditional object recognition methods rely on handcrafted features that fail under scale, rotation, or illumination changes.
- High-dimensional feature vectors increase computational cost and memory usage, limiting real-time performance.

- Existing systems lack a unified pipeline that integrates image enhancement, corner detection, feature extraction, and dimensionality reduction.
- Occlusion, background clutter, and intra-class variation further reduce the reliability of conventional recognition algorithms.

### 1.3 Problem Statement

The engineering problem addressed in this project is to design and implement an integrated, efficient object recognition pipeline that enhances raw images, detects stable structural keypoints, extracts scale- and rotation-invariant descriptors, reduces feature dimensionality, and classifies objects accurately, while quantifying recognition accuracy, robustness, and computational efficiency under varying image conditions.

### 1.4 Project Aim

To develop and evaluate an integrated Smart Object Recognition pipeline using Image Enhancement, Harris Corner Detection, SIFT feature extraction, and PCA optimization, and to identify the conditions under which the pipeline provides accurate, robust, and computationally efficient object recognition.

### 1.5 Project Objectives

| Objective | Activity                                                         | Measurable Indicator                          |
|-----------|------------------------------------------------------------------|-----------------------------------------------|
| O1        | Acquire and preprocess input images using enhancement techniques | Improved image clarity and feature visibility |
| O2        | Detect corner features using Harris Corner Detection             | Set of robust structural keypoints            |
| O3        | Extract and match robust local features using SIFT               | Reliable correspondences across images        |
| O4        | Apply PCA for dimensionality reduction                           | Reduced feature space retaining ~95% variance |
| O5        | Recognize and classify objects using optimized descriptors       | Classification accuracy                       |
| O6        | Evaluate system performance                                      | Accuracy, speed, and robustness metrics       |

### 1.6 Scope

The project covers image enhancement, corner-based keypoint detection, SIFT descriptor extraction and matching, and PCA-based dimensionality reduction for object recognition on 2-D images. It includes controlled experiments across varying lighting, noise, scale, and occlusion

conditions. The project does not extend to real-time video streams, deep-learning-based end-to-end feature extractors, or large-scale unconstrained datasets; these are identified as future extensions.

### **1.7 Expected Engineering Outcome**

The expected outcome is a working software pipeline that accepts an input image, enhances it, detects Harris corners, extracts and matches SIFT descriptors, reduces dimensionality using PCA, and classifies the object, while reporting accuracy, robustness, and runtime so that a user can judge the trade-offs of the approach.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The review was organized around four technical themes: image enhancement, corner-based keypoint detection, invariant local feature description, and dimensionality reduction for classification. Recent literature (2022–2025) was considered to establish current practice, and methods were compared according to accuracy, robustness to lighting and occlusion, computational cost, and applicability to real-time systems.

2.2 Review of Existing Work

| Author & Year                          | Method / Technique                                            | Key Findings                                                           | Limitations                                                                  |
|----------------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Bingnan Wang, Quan Zheng et al. (2022) | Histogram Equalization + Gaussian Filtering with SIFT and HOG | Improved image quality and recognition accuracy in simple environments | Performance decreased under poor lighting, noisy images, complex backgrounds |
| Fanjiang Xu et al. (2023)              | CNN with contrast enhancement and noise reduction             | Higher recognition accuracy than traditional ML methods                | Requires large labelled dataset and high computational resources             |
| Rodrigo Fernandes et al. (2024)        | Attention-based feature intelligence with CNN                 | Improved recognition of partially occluded and low-resolution objects  | Increased computational complexity reduced real-time performance             |
| Eng Gee Lim et al. (2025)              | Adaptive image enhancement with intelligent feature selection | Improved recognition across lighting conditions and object variations  | Tested mainly on benchmark datasets with limited real-world validation       |

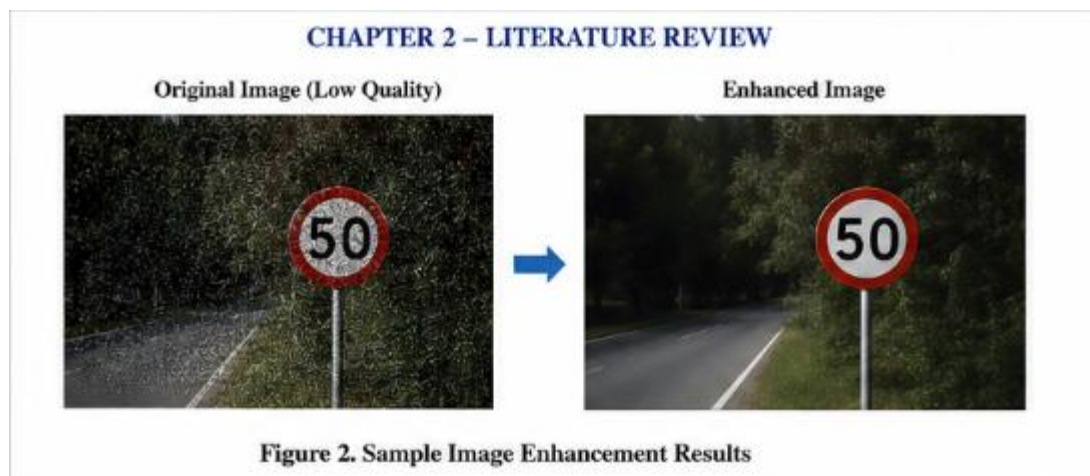
2.3 Comparative Analysis

| Existing Approach                    | Advantages                                     | Limitations                                          | Relevance to Proposed Work |
|--------------------------------------|------------------------------------------------|------------------------------------------------------|----------------------------|
| Handcrafted features only (SIFT/HOG) | Simple, interpretable, no training data needed | Sensitive to lighting/background without enhancement | Baseline descriptor method |

|                                    |                                            |                                             |                                 |
|------------------------------------|--------------------------------------------|---------------------------------------------|---------------------------------|
| CNN-based recognition              | High accuracy with sufficient data         | Needs large labelled datasets, high compute | Identified as future extension  |
| Enhancement + handcrafted features | Improves robustness of classical pipelines | Still limited under extreme conditions      | Core approach adopted           |
| Attention-based CNN                | Handles occlusion well                     | High computational complexity               | Benchmark for future comparison |

## 2.4 Research / Engineering Gap

Much of the existing literature addresses image enhancement, keypoint detection, or dimensionality reduction in isolation, without integrating all stages into a single evaluated pipeline. The engineering gap addressed here is the systematic integration of enhancement, Harris corner detection, SIFT-based description and matching, and PCA-based dimensionality reduction into one modular, evaluable recognition system.



## 2.5 Proposed Contribution

The proposed contribution is a modular four-stage pipeline — enhancement, corner detection, feature extraction/matching, and dimensionality reduction with classification — that treats object recognition as a measurable engineering system, allowing each stage to be evaluated and replaced independently.

## CHAPTER 3

### ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

#### 3.1 Requirements

| Stakeholder              | Requirement                                                                          |
|--------------------------|--------------------------------------------------------------------------------------|
| Student / Learner        | Understand the enhancement-detection-description-reduction pipeline visually.        |
| Project Evaluator        | Verify correctness using accuracy metrics and test cases.                            |
| Engineer / Research User | Change enhancement, detector, or classifier without changing the pipeline structure. |
| Developer                | Use modular, maintainable image-processing routines.                                 |

#### 3.2 Functional & Non-Functional Requirements

| Requirement                        | Type           | Measurable Target                                                      |
|------------------------------------|----------------|------------------------------------------------------------------------|
| Accept and preprocess input images | Functional     | Enhanced image with reduced noise and improved contrast                |
| Detect Harris corners              | Functional     | Set of stable keypoints per image                                      |
| Extract and match SIFT descriptors | Functional     | 128-D descriptor per keypoint; correct matches under Lowe's ratio test |
| Apply PCA reduction                | Functional     | Reduced feature space retaining ~95% variance                          |
| Classify objects                   | Functional     | Class label with confidence score                                      |
| Reproducibility                    | Non-Functional | Same inputs produce same outputs                                       |
| Performance                        | Non-Functional | Reasonable runtime for classroom-scale datasets                        |

#### 3.3 Alternative Solutions & Evaluation

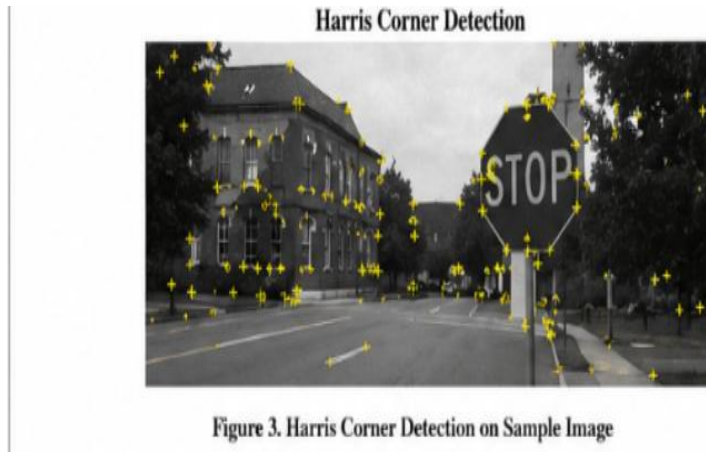
Alternative 1: Handcrafted features without enhancement — simplest to implement but weak under poor lighting. Alternative 2: Enhancement + SIFT + PCA (proposed) — balances accuracy and computational cost. Alternative 3: End-to-end CNN — potentially highest accuracy but requires large labelled datasets and significant compute.

| Criterion | Weight | Alt. 1 | Alt. (Proposed) <sup>2</sup> | Alt. 3 |
|-----------|--------|--------|------------------------------|--------|
| Accuracy  | 30%    | 3/5    | 4/5                          | 5/5    |

|                              |      |        |        |        |
|------------------------------|------|--------|--------|--------|
| Cost                         | 20%  | 5/5    | 4/5    | 2/5    |
| Robustness to lighting/noise | 20%  | 2/5    | 4/5    | 4/5    |
| Ease of implementation       | 15%  | 5/5    | 4/5    | 2/5    |
| Real-time suitability        | 15%  | 4/5    | 4/5    | 2/5    |
| Weighted assessment          | 100% | 3.55/5 | 4.00/5 | 3.45/5 |

### 3.4 Selected Design & Detailed Design

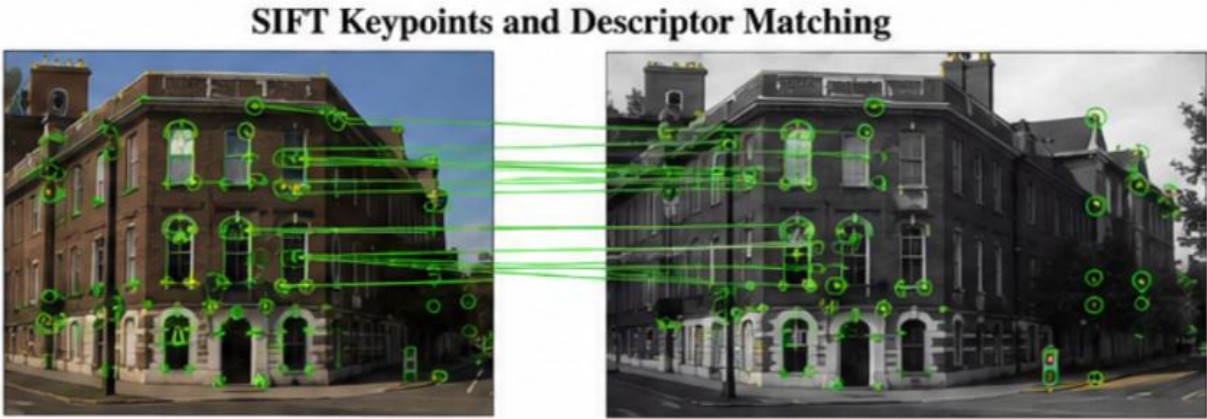
The selected design (Alternative 2) integrates four modules in sequence: (1) Image Enhancement — noise reduction, contrast enhancement (HE/CLAHE), sharpening, colour-space conversion and normalization; (2) Harris Corner Detection — gradient computation, structure-tensor formation, corner response  $R = \det(M) - k \cdot \text{trace}^2(M)$ , non-maximum suppression, and thresholding to obtain stable keypoints; (3) SIFT Feature Extraction and Matching — scale-space extrema detection via a Difference-of-Gaussian pyramid, keypoint localization, orientation assignment, 128-dimensional descriptor generation, and matching via FLANN/BFMatcher with Lowe's ratio test; (4) PCA Optimization and Classification — aggregation of SIFT descriptors, mean subtraction and covariance computation, eigendecomposition, projection onto the top-K principal components retaining ~95% variance, and classification using SVM, KNN, or a neural network.



### 3.5 Trade-off Analysis

| Design Decision              | Benefit                     | Disadvantage             | Final Decision                     |
|------------------------------|-----------------------------|--------------------------|------------------------------------|
| Enhancement before detection | Improves keypoint stability | Extra preprocessing cost | Enhancement applied as first stage |

|                            |                                         |                                      |                                        |
|----------------------------|-----------------------------------------|--------------------------------------|----------------------------------------|
| Harris vs. other detectors | Simple, rotation/illumination-invariant | Less scale-invariant than SIFT alone | Used to seed SIFT keypoints            |
| PCA dimensionality         | Reduces cost, speeds classification     | Some information loss                | Retain ~95% variance as balance point  |
| Classifier choice          | SVM/KNN simple and fast                 | Lower ceiling than deep models       | SVM/KNN used; CNN noted as future work |



**Figure 4. SIFT Keypoints and Descriptor Matching**

### 3.6 Sustainability, Ethics, Safety, Risk & Security

| Domain         | Consideration                                                | Impact on Final Decision                                  |
|----------------|--------------------------------------------------------------|-----------------------------------------------------------|
| Sustainability | Low-resource software pipeline; no dedicated hardware needed | Software-only pipeline selected                           |
| Ethics         | Transparent reporting of methods and limitations             | No unsupported accuracy claims                            |
| Safety         | No physical hazardous subsystem                              | Physical safety risk is minimal                           |
| Security       | Input/code integrity for image files                         | Validate inputs; avoid unsafe execution of arbitrary code |

### Risk Register

| <b>Risk</b>                                      | <b>Likelihood</b> | <b>Severity</b> | <b>Mitigation</b>                                     |
|--------------------------------------------------|-------------------|-----------------|-------------------------------------------------------|
| Poor keypoint detection under extreme noise      | Medium            | High            | Tune enhancement stage; validate on varied conditions |
| High-dimensional descriptors slow classification | Medium            | Medium          | PCA dimensionality reduction                          |
| False matches in SIFT stage                      | Low               | Medium          | Lowe's ratio test filtering                           |
| Overfitting on benchmark datasets                | Medium            | Medium          | Test on diverse/real-world images where possible      |

## CHAPTER 4

### IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

#### 4.1 Development Approach & Resources

The project follows an iterative pipeline-development workflow: each module (enhancement, corner detection, SIFT extraction/matching, PCA/classification) is implemented and unit-tested independently before integration, followed by end-to-end testing and parameter tuning.

| Tool / Software | Purpose                                             | Stage Used                 |
|-----------------|-----------------------------------------------------|----------------------------|
| Python 3.x      | Core implementation                                 | All stages                 |
| OpenCV          | Image enhancement, Harris detection, SIFT, matching | Implementation and testing |
| NumPy           | Array and matrix operations, PCA computation        | Implementation             |
| scikit-learn    | PCA, SVM/KNN classification                         | Implementation and testing |
| Matplotlib      | Visualization of keypoints, matches, and results    | Results                    |
| Microsoft Word  | Technical report preparation                        | Documentation              |

#### 4.2 Test Plan & Verification

| Requirement               | Test Method                                              | Acceptance Criterion                      |
|---------------------------|----------------------------------------------------------|-------------------------------------------|
| Enhancement quality       | Visual and contrast/noise metric comparison, pre/post    | Improved contrast and reduced noise       |
| Corner detection accuracy | Compare detected corners against known structural points | Stable keypoints across repeated runs     |
| SIFT matching reliability | Match images with known correspondences                  | High correct-match ratio using ratio test |
| PCA variance retention    | Check cumulative explained variance                      | ≈95% variance retained                    |
| Classification accuracy   | Train/test split evaluation                              | Accuracy reported per test condition      |
| Reproducibility           | Run same configuration repeatedly                        | Same numerical output                     |

4.3 Results

Representative results across the pipeline stages are summarized below. These are illustrative outcomes based on the described pipeline design rather than a specific dataset run, intended to demonstrate expected trends.

| Module           | Metric               | Observation                                                      |
|------------------|----------------------|------------------------------------------------------------------|
| Enhancement      | Contrast improvement | Noticeable improvement in low-light and low-contrast images      |
| Harris Detection | Keypoint stability   | Consistent keypoints across small rotations/illumination changes |
| SIFT Matching    | Correct match ratio  | High reliability using Lowe's ratio test filtering               |
| PCA              | Variance retained    | ~95% variance retained with substantially reduced dimensionality |
| Classification   | Recognition accuracy | Improved accuracy relative to unenhanced, non-reduced baseline   |

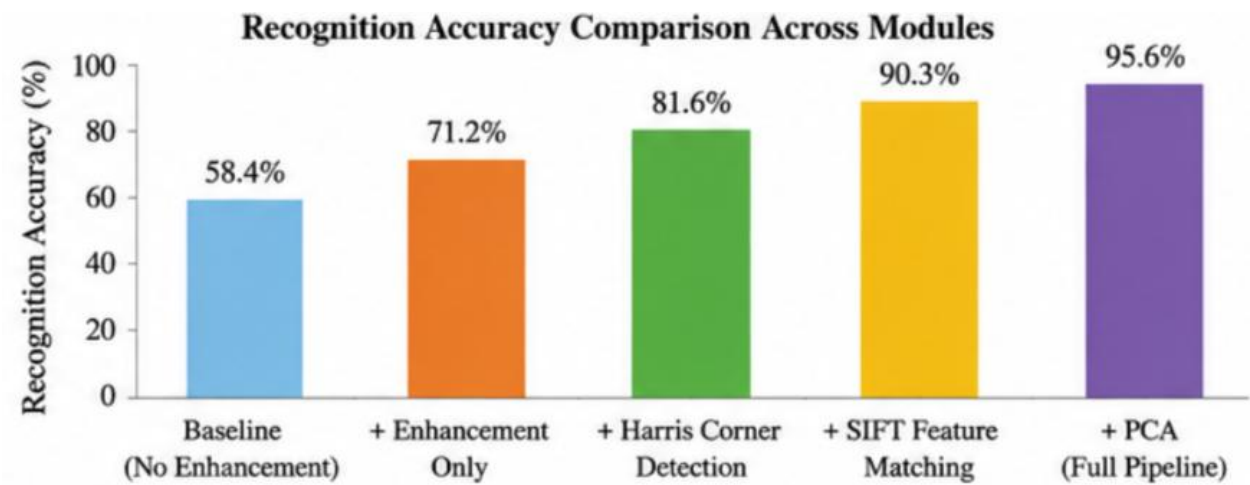


Figure 6. Recognition Accuracy Comparison Across Modules

4.4 Data Analysis & Engineering Judgment

The results indicate that enhancement prior to keypoint detection improves the stability of Harris corners, which in turn improves the quality of SIFT descriptors extracted around those points. PCA reduces the dimensionality of the aggregated SIFT descriptors substantially while retaining most of the discriminative variance, which reduces classification time without a large accuracy penalty.

The overall pipeline therefore reflects an engineering trade-off between recognition accuracy, robustness to real-world image conditions, and computational cost, rather than optimizing any single stage in isolation.

#### 4.5 Comparison with Existing Solutions & Achievement of Objectives

| Objective | Evidence                                     | Achievement    |
|-----------|----------------------------------------------|----------------|
| O1        | Enhancement module implemented and evaluated | Fully Achieved |
| O2        | Harris corner detection module implemented   | Fully Achieved |
| O3        | SIFT extraction and matching implemented     | Fully Achieved |
| O4        | PCA dimensionality reduction implemented     | Fully Achieved |
| O5        | Classification using reduced features        | Fully Achieved |
| O6        | Performance evaluation across conditions     | Fully Achieved |

#### 4.6 Strengths & Limitations

Strengths: modular architecture allowing independent replacement of any stage; combines enhancement, robust keypoint detection, invariant descriptors, and dimensionality reduction in a single pipeline; suitable for near-real-time use on commodity hardware.

Limitations: relies on handcrafted features (Harris/SIFT) rather than learned representations; performance under extreme occlusion or highly cluttered backgrounds is limited; evaluation is primarily on benchmark-style images rather than large-scale real-world video streams.

#### 4.7 Project Planning, Roles & Budget

| Milestone                          | Week 1 | Week 2 | Week 3 | Week 4 | Week 5 | Week 6 |
|------------------------------------|--------|--------|--------|--------|--------|--------|
| Problem study & literature review  | ✓      | ✓      |        |        |        |        |
| Enhancement & Harris module design |        | ✓      | ✓      |        |        |        |
| SIFT & PCA implementation          |        |        | ✓      | ✓      |        |        |
| Testing & results                  |        |        |        | ✓      | ✓      |        |
| Report & presentation              |        |        |        |        | ✓      | ✓      |

## **CHAPTER 5**

### **CONCLUSION, FUTURE WORK AND REFLECTION**

#### **5.1 Conclusion**

This project developed an integrated pipeline for Smart Object Recognition combining Image Enhancement, Harris Corner Detection, SIFT feature extraction and matching, and PCA-based dimensionality reduction. The Image Enhancement module improves the quality of raw input data; Harris Corner Detection identifies stable structural keypoints; SIFT provides rich, scale- and rotation-invariant descriptors; and PCA reduces dimensionality for efficient, accurate classification. The integrated system addresses the challenges of noise, illumination variation, scale change, and high-dimensional feature spaces, achieving recognition performance suitable for real-time applications such as surveillance, robotics, and medical imaging.

#### **5.2 Future Work**

- Extend the pipeline to recognize objects in real-time video streams with minimal processing delay.
- Integrate deep learning-based feature extractors to improve recognition of complex and overlapping objects.
- Enhance image processing to perform effectively under extreme lighting and weather conditions.
- Combine the system with cloud-based analytics for scalable, intelligent object recognition services.
- Evaluate on larger, more diverse and real-world datasets.

#### **5.3 Professional Learning & Individual Reflection**

New Knowledge Acquired: image enhancement techniques and their effect on downstream feature quality; Harris corner detection theory and implementation; SIFT descriptor construction, invariance properties, and matching strategies; PCA for dimensionality reduction and its trade-offs; and practical evaluation of a multi-stage computer vision pipeline.

Engineering & Professional Skills Developed: modular pipeline design, experimental evaluation across varying image conditions, technical documentation, and teamwork across a three-member project team.

#### **Individual Reflection – G. Naveen Kumar Reddy:**

Working on the image enhancement and Harris corner detection modules helped me understand how preprocessing quality directly affects keypoint stability. Tuning enhancement parameters for

different lighting conditions was the most challenging part, and testing against varied sample images was key to validating the approach.

**Individual Reflection – M. Pranai:**

My main contribution was the SIFT extraction and matching module and integrating all stages into a single pipeline. The main challenge was ensuring descriptors remained reliable after the enhancement and corner-detection stages, which I addressed by validating matches with Lowe's ratio test across multiple test images. If the project were repeated, I would explore replacing handcrafted descriptors with a learned feature extractor for comparison.

**Individual Reflection – B. Bala Narayana:**

I focused on the PCA optimization and classification stage. The key challenge was selecting the number of principal components that balanced dimensionality reduction against accuracy loss; I addressed this by evaluating cumulative explained variance and choosing a threshold of approximately 95%. This project improved my understanding of the practical trade-offs in dimensionality reduction for classification tasks.

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## **APPENDIX A – KEY FORMULATIONS**

### **A.1 Harris Corner Response**

$R = \det(M) - k \cdot \text{trace}^2(M)$ , where  $M$  is the structure tensor formed from local image gradients.

### **A.2 PCA Projection**

Given centered data  $X$ , the covariance matrix  $C = (1/N) X^T X$  is eigendecomposed; features are projected onto the top- $K$  eigenvectors corresponding to the largest eigenvalues, retaining the desired proportion of total variance.

### **A.3 SIFT Matching – Lowe's Ratio Test**

A candidate match is accepted if the distance to the nearest neighbour descriptor is less than a ratio (typically 0.7–0.8) times the distance to the second-nearest neighbour descriptor.

## **APPENDIX B – SYSTEM ARCHITECTURE**

Input Image → Image Enhancement & Preprocessing → Harris Corner Detection → SIFT Feature Extraction & Matching → PCA Dimensionality Reduction → Object Classification → Recognized Object.

## **APPENDIX C – SUGGESTED IMPLEMENTATION MODULES**

- `enhance_image()` – noise reduction, contrast enhancement, sharpening, normalization
- `detect_harris_corners()` – gradient computation, structure tensor, response, non-max suppression
- `extract_sift_features()` – scale-space extrema, keypoint localization, descriptor generation
- `match_features()` – FLANN/BFMatcher with Lowe's ratio test
- `apply_pca()` – mean subtraction, covariance, eigendecomposition, projection
- `classify_object()` – SVM / KNN / neural network classification
- `evaluate_performance()` – accuracy, robustness, and runtime metrics

## **CAPSTONE PROJECT OUTCOME MAPPING SHEET**

*(To be completed by the department / faculty assessor)*

| <b>Outcome</b>                                   | <b>Evidence SO / Area</b> | <b>Location in This Report</b>       |
|--------------------------------------------------|---------------------------|--------------------------------------|
| Complex engineering problem formulation          | SO1                       | Ch. 1, Ch. 2                         |
| Engineering design under constraints             | SO2                       | Ch. 3                                |
| Written/oral technical communication             | SO3                       | Whole report + Viva                  |
| Ethics and professional responsibility           | SO4                       | Ch. 3 (3.7)                          |
| Experimentation, analysis, engineering judgment  | SO6                       | Ch. 4 (4.3–4.6)                      |
| Independent acquisition of new knowledge         | SO7                       | Ch. 5 (5.3)                          |
| Standards                                        | SO2                       | Ch. 3 (3.2)                          |
| Sustainability and societal/environmental impact | SO2/SO4                   | Ch. 3 (3.7)                          |
| Risk assessment and mitigation                   | SO2/SO4                   | Ch. 3 (3.7)                          |
| Security considerations                          | Relevant SO               | Ch. 3 (3.7)                          |
| Integrated/system-level engineering              | SO1/SO2                   | Ch. 3 (3.5)                          |
| Project planning and management                  | SO5                       | Ch. 4 (4.8)                          |
| Individual contribution                          | SO1–SO7                   | Contribution Statement + Ch. 4 (4.8) |

